

Common Problems and Fixes for Using the Cluster

1.

- a. **Problem - Missing Python Packages:** ModuleNotFoundError: | No module named ____
- b. **Cause:** The given module in the error was not installed properly into a virtual environment used during execution.
- c. **Fix:** Update your pip install statement to include the modules not found. Make sure to run this with the virtual environment used during execution activated.

2.

- a. **Problem - GCC / Arrow Module Problems:**
Deactivate the environment
module load gcc arrow/x.y.z
- b. **Cause:** Because packages are loaded off a wheel, some Python libraries require the Arrow C++ libraries. If these modules are not loaded correctly, pip cannot install packages.
- c. **Fix:** Deactivate your virtual environment (deactivate), purge old modules (module purge), load the new modules (module load gcc arrow/x.y.z), reactivate the virtual environment, and run the slurm script again. NOTE: The x.y.z following arrow should be replaced with the version required. Find this in the error message (search for something like (from pyarrow>=21.0.0->datasets)).

3.

- a. **Problem - CUDA Problems:** AssertionError: CUDA is required
- b. **Cause:** Torch couldn't see the GPU required to run your script on one of the cluster's GPUs
- c. **Fix:** Request a GPU in slurm (--gres=gpu:<GPU MODEL>:<NUM OF REQUESTED GPUS>). Make sure CUDA-enabled PyTorch is installed (module load cuda may be required).

4.

- a. **Problem - Slurm Job Pending Forever:** ReqNodeNotAvail | UnavailableNodes| Nodes required for job are DOWN DRAINED or RESERVED for jobs in higher priority partitions
- b. **Cause:** The cluster does not have nodes available to allocate for your job. The cluster could be down for maintenance or just really busy.
- c. **Fix:** Either wait, remove unnecessary resource allocations (ex. If only 20G of memory is required, don't request more), or switch to another cluster.

5.

- a. **Problem - Out of GPU Memory:** CUDA out of memory
- b. **Cause:** The job on the GPU tried to get too much memory, leading to this error
- c. **Fix:** For training AI, decrease batch size, increase gradient accumulation, enable gradient checkpointing, use bf16 or fp16, implement QLoRA instead of full fine tuning, or look at running on more GPUs (use DeepSpeed for AllianceCan clusters). For other types of operations on the cluster, look at ways of decreasing memory requirements, or increasing the number of GPUs requested (may require sharding the job across multiple GPUs/nodes).

6.

- a. **Problem - Wrong GPU Requested:** GPU job without specifying a GPU type
- b. **Cause:** A GPU type was not properly requested in the slurm script
- c. **Fix:** In the slurm script, instead of using something like #SBATCH --gres=gpu:1 use #SBATCH --gres=gpu:h100:1 (or whatever your cluster requires).

7.

- a. **Problem - SSH Agent Forwarding not Working:** Username/password authentication failed
- b. **Cause:** SSH agent forwarding was not set up right, or the ssh agent tag was not used in the SSH command
- c. **Fix:** Ensure the setup correctly executed (use `ssh -T git@github.com`, should print out a welcome message), and make sure you login to ssh with the -A tag (`ssh -A yourusername@clusterdomain.ca`)

8.

- a. **Problem:** Wrong file paths
- b. **Cause:** The cluster has a different directory structure than a local machine
- c. **Fix:** Switch file paths so that they use `os.path.join(...)` instead of the paths previously used, or use relative paths that are specific to the cluster.

9.

- a. **Problem:** Out of disk space
- b. **Cause:** The cluster directories are to be used for different purposes.
Writing/reading to the wrong directory can make the cluster exceed disk quotas.
- c. **Fix:** Figure out where the disk space is running out with `diskusage_report`.
Change specified directories to make use of areas such as `scratch/`

10.

- a. **Problem:** Hugging Face cache is exceeding disk quotas
- b. **Cause:** When AI models are loaded from Hugging Face, they load into the Hugging Face cache. If these models are very large, they can exceed disk space for certain directories.
- c. **Fix:** Switch the directory of the Hugging Face Cache (`export HF_HOME="/path/to/your/custom/cache"`)

11.

- a. **Problem:** Accidentally tracking large files on GitHub
- b. **Cause:** If some files are not explicitly added to the git ignore file, they will be tracked in GitHub, which can cause issues if the files are very large.
- c. **Fix:** To keep them locally while removing them from the GitHub repository, use `git rm --cached -r yourfile`. It is best practice to move large files between your computer and the cluster with `scp` or a similar file transfer protocol.